



THE UNIVERSITY *of* EDINBURGH
Digital Research Services

2024 DRS AMBASSADOR Internship Scheme

Students' approach to the use of generative AI.

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Abstract

In this project, our aim was to understand how students in the Digital Business course (n=192) used ChatGPT to aid with a coursework exercise. The students had to write a business proposal for a competition, where the most original proposal would win. First the students wrote proposals without the use of generative AI (task 1), and then refined them using ChatGPT (task 2). The project host defined originality of a proposal as the inverse of similarity to all other proposals.

As the Digital Research Services Ambassador in this project, I helped the team evaluate the originality of student proposals by using a variety of computer calculated metrics. Over the course of the project, both I and the hosts learned a lot about measures for text similarity – we tried orthographic (character-level) similarity, sequence (string) similarity, and cosine similarity with static and contextual embeddings. Ultimately, we decided to use an average of multiple similarity measures and averaging between cosine distance to all other proposals and 10 closest proposals within the same task.

I found a few curious patterns, for example, task 1 texts (without ChatGPT) were distinctly dissimilar from task 2 texts (with ChatGPT) and a short passage of mock irrelevant data had high similarity to submissions that were inappropriate for the set tasks. My calculations, code, documentation, and analysis will form part of the larger project discussion about students' use of generative AI.



1. PROJECT OVERVIEW

1.1. Original Host Project Proposal

The intern would participate in a project that aims to understand how students use generative AI and what the outcomes are of this use. The project involves an in-class experiment, where students perform an in-class exercise, using generative AI. The research group wants to compare the prompts for the tool as well as the tool outputs and the students' final delivery in terms of similarity. They aim to provide robust analyses that will inform education professionals and institutions to help them to understand students' approach to, use of, and understanding of AI. This will allow Universities to situate their AI policy and guidance within a deep understanding of student perspectives and practices. The study will also enable a discussion at a societal level regarding the use of generative AI in educational contexts.

1.2. Original Host Project Detail

We have two objectives for this project. We will focus on the first one and do the second if we have time.

Students have performed two tasks, one with the support of ChatGPT and one without. The task consisted of creating a business idea.

We want to do two things with this data.

1. Calculate within-student text similarity (i.e., text similarity between the first and the second tasks).
2. Calculate cross-student text similarity (i.e., text similarity between each text and the bulk of all the other texts).

I am sending you a list of references for the measures of similarity. For the first objective (our focus), I believe the first text cited below is very useful.

I have collected the articles and the corresponding replication data here, so that you can have a look to see how complex the task is. I have sent a separate message to you.

If you want to consider task 2, you may want to either adapt the same technique or have a look at the other articles below.

Guzman, J., & Li, A. (2023). Measuring founding strategy. Management Science, 69(1), 101-118. <https://doi.org/10.1287/mnsc.2022.4369>

NLP - "doc2vec" algorithm to generate vectors representing textual content.

Similarity - cosine similarity between the respective doc2vec-generated vectors of a pair of texts.

Barlow, M. A., Verhaal, J. C., & Angus, R. W. (2019). Optimal distinctiveness, strategic categorization, and product market entry. *Strategic Management Journal*, 40(12), 1219-1242. <https://doi.org/10.1002/smj.3019>

Technique summary: Relevant to us - Exemplar Similarity: This measure evaluates the similarity between the text description of a new app and the descriptions of the top-performing (exemplar) apps within the same category. Exemplars are considered the most successful or salient products in the category. The similarity is quantified using cosine similarity between vector representations of the app descriptions, where higher values indicate greater similarity.

Arts, S., Cassiman, B., & Gomez, J. C. (2017). Text matching to measure patent similarity. *Strategic Management Journal*, 39(1), 62-84. <https://doi.org/10.1002/smj.2699>

Technique summary: Generates a list of keywords of each patent representing the technical content of patents. To measure the similarity between any two patents, the Jaccard index is used. This index is calculated by dividing the number of unique keywords in the intersection (shared keywords) by the number of unique keywords in the union of the two patent documents. The resulting measure ranges from zero (no similarity) to one (identical content).

1.3. Ambassador Project Plan

In a previous study modelling creativity of Scratch projects created by children through an automatic measure, originality was one of the elements of creativity (Kovalkov et al., 2021). The authors measured originality by distance to a sample solution. This approach worked because there were pre-defined categories of projects that could have sample solutions. In our case, students had an open task with lots of freedom what to write about, hence there could be no sample solution. Instead, using average distance to all other texts seems like a sensible measure of originality.

Information Retrieval techniques for determining similarity between texts range in granularity from considering individual characters to whole documents at a time.

At character level, there are metrics like:

- Hamming distance
- Damerau-Levenshtein distance
- Jaro-Winkler distance

These capture similarity in spelling. Such string metrics can handle typos and misspellings (but it also means words like “good” and “food” are highly similar).

For sequences, there are metrics like:

- Jaccard index
- Dice-Sørensen coefficient
- Longest common subsequence
- Longest common substring

These handle common words, rather than individual characters. But they don't incorporate anything about semantic meanings. Pre-processing can help to match words with the same base, like "love" and "loving", but otherwise, words like "sky", "cloud", and "cat" are considered equally different from each other, even though we might expect "sky" and "cloud" to be a bit more similar to each other than to "cat". Bašić et al. (2023), in a study analysing student essays written with assistance from ChatGPT 3, used Jaccard index to measure similarity.

There are static embedding approaches, for example:

- Term Frequency – Inverse Document Frequency (TF-IDF)
- Doc2Vec (Google, 2014 ; <https://arxiv.org/abs/1405.4053>)

These capture similarity in semantic meaning, but each word always maps to the same embedding, which doesn't account for words with multiple meanings that depend on context.

Contextual embeddings, e.g.:

- Universal Sentence Encoder (Google, 2018 ; <https://arxiv.org/abs/1803.11175>)
- SBERT and sentence transformers (UKP-TUDA, 2019 ; <https://arxiv.org/pdf/1908.10084> ; <https://www.sbert.net/>)

These capture semantic meanings in context and so have the best representations of small paragraphs. However, too much information can degrade performance, and contextual embeddings can be worse for global comparison of large texts than basic approaches.

To calculate similarity between embeddings (static or contextual) several metrics can be used, such as:

- Euclidean distance
- Cosine similarity
- Dot score

For normalised vectors, all 3 metrics have a direct relationship with each other, so specific choice is not too important. Out of them, cosine similarity (or dot score, which is equivalent for normalised vectors) is most commonly used.

Some basic natural language pre-processing techniques can help with all the above approaches, like removing punctuation, lowercasing all text, removing common stop words, and stemming or lemmatising (turning words into their base forms).



I propose computing similarity using several complementary approaches as they capture different aspects and each have their pros/cons. For example, we might pick Damerau-Levenshtein distance, Jaccard index, TF-IDF, and a current embeddings model with sentence-transformers. If there are disparities between them, we can then investigate further.

The [MTEB Leaderboard - a Hugging Face Space by mteb](#) has up-to-date rankings of embeddings models. [Alibaba-NLP/gte-base-en-v1.5 · Hugging Face](#) (Institute for Intelligent Computing, Alibaba Group, 2023; <https://arxiv.org/pdf/2308.03281>) were leading in average overall performance in embedding models with less than 250 million parameters at the time of writing this initial proposal.

Calculating document similarities with these different approaches is fairly easy – there are a number of tutorials and libraries with ready-made code that just needs to be adapted a little, such as: [Ultimate Guide To Text Similarity With Python - NewsCatcher \(newscatcherapi.com\)](#) and [Semantic Textual Similarity — Sentence-Transformers documentation \(sbert.net\)](#). The most difficult part will be inferring causal connections and drawing conclusions about how similarity scores are related to student behaviour.

The approach is basically the same when comparing within students and between students pairwise. But for between students, we may also want to try clustering and topic allocation techniques, like K-means clustering and Latent Dirichlet Allocation.

Some other alternative investigation approaches could be:

- Manually labelling key features in texts, graphing a network of correlations, and drawing inferences from those
- Educators manually labelling features in texts like cognitive processes from Bloom's taxonomy where relevant (like [Assessing the Quality of Student-Generated Short Answer Questions Using GPT-3 | SpringerLink](#))
- Or using Turnitin for similarity (like [OpenAI ChatGPT Generated Content and Similarity Index: A study of selected terms from the Library & Information Science \(LIS\) \(semanticsscholar.org\)](#))

1.4. Data Protection and Ethics Considerations

As the project involves handling student data, we must be careful to work within the constraints of the original approved ethics application. Relevant aspects:

- All data is pseudonymised
- Students have the right to withdraw from the study (so it should be possible to easily re-run my code with modified data if some of it is removed)
- Data must not leave university premises / cloud services

For testing, I will work on my own laptop with an anonymised subsample of the original dataset. Once ready, we can then run the code on the full dataset on university infrastructure.

1.5. Dataset

The submitted business proposals were written in response to the following tasks:

- Task 1: "You have just joined a team of three entrepreneurs who will start a business together. All the team has decided so far is that you will start a digital platform that facilitates transactions or interactions among users. Your team wants to enter a competition where you can get £100,000 in seed funding. Your role is to write a business proposal with the potential to win the competition. In this competition, business proposals will be evaluated according to their creativity."
- Task 2: "Now you have the opportunity to consult ChatGPT before submitting your final proposal. You will consult ChatGPT using at least 2 prompts*, in the ways you consider the best to successfully finish this task. (*Prompts are instructions given to artificial intelligence.) Once you are satisfied, you will choose one of the three options below: Please choose the option that gives you the highest chance of winning the competition."

Due to the data protection considerations, I cannot include example of the texts in this report, but for some descriptive statistics: we had 192 paired texts submitted by students (for task 1 and task 2) and the texts ranged in length from 144 words to 16600 words – so the dataset included both very short and very long texts. The median text length for task 1 was 966.5 words and for task 2 – 1661.5 words.

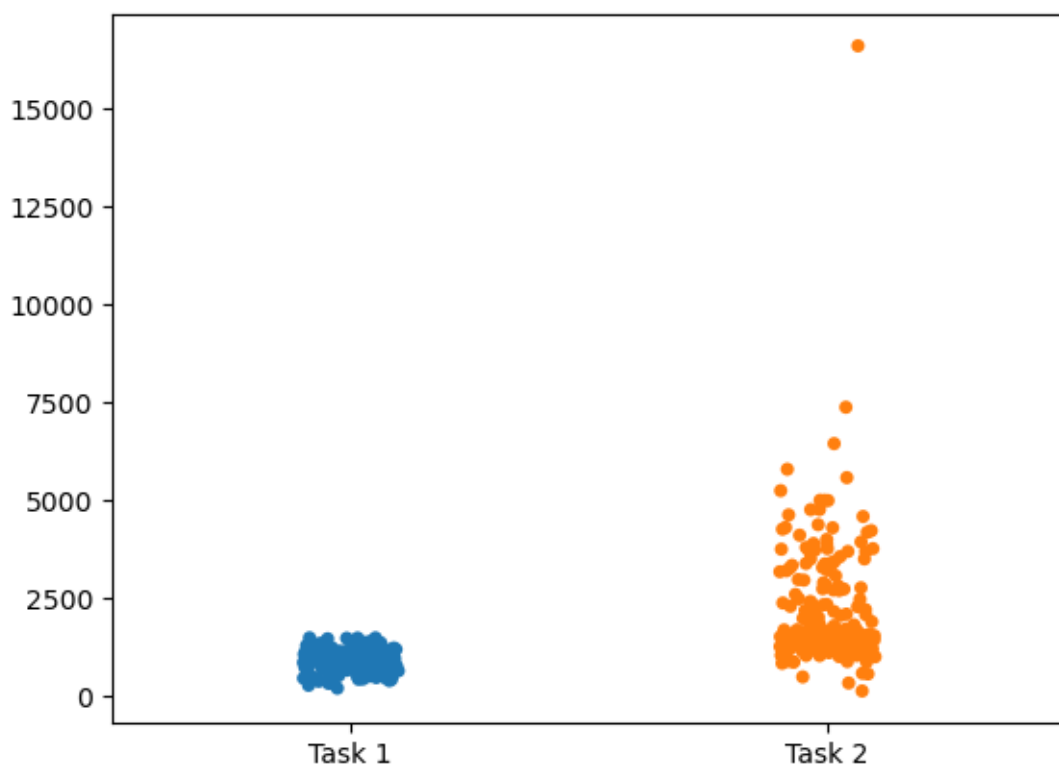


Figure 1: Text lengths per task

To pre-process these raw texts, I used the following steps:

1. Remove all punctuation
2. Lowercase all the text
3. Split up the text into tokens
4. Remove numbers (but not words that contain numbers)
5. Remove stopwords (common English words that do not add unique meaning)
6. Remove tokens that are only 1 character long
7. Lemmatise each token using WordNet

These steps are standard in natural language processing and mostly built-in with the [NLTK :: Natural Language Toolkit](#) library.

1.6. Similarity metrics

To calculate text similarity, I used the following metrics:

Metric	Metric type	Data	Range
Jaro-Winkler similarity	Character-level	Pre-processed texts	[0,1]
Euclidian similarity	Character-level	Pre-processed texts	[0,1]
Compression similarity	Subword-level	Pre-processed texts	[0,1]
Jaccard similarity	Word-level	Pre-processed texts	[0,1]
Jaccard similarity	Bigram-level	Pre-processed text bigrams	[0,1]
Jaccard similarity	Trigram-level	Pre-processed text trigrams	[0,1]
Cosine similarity	Static embeddings	Term-Frequency Inverse-Document-Frequency representations of pre-processed texts	[0,1]
Cosine similarity	Static embeddings	Okapi BM25 representations of pre-processed texts	[0,1]
Cosine similarity	Static embeddings	Doc2Vec model (trained on pre-processed texts) embeddings of pre-processed texts	[-1,1]
Cosine similarity	Static embeddings	Doc2Vec model (trained on website dataset from Guzman and Li (2023)) embeddings of pre-processed texts	[-1,1]
Cosine similarity	Contextual embeddings	Alibaba-NLP/gte-large-en-v1.5 model embeddings of raw texts	[-1,1]
Cosine similarity	Contextual embeddings	mixedbread-ai/mxbai-embed-large-v1 model embeddings of raw texts	[-1,1]
Cosine similarity	Contextual embeddings	Alibaba-NLP/gte-Qwen2-7B-instruct model embeddings of raw texts	[-1,1]

Table 1: Similarity metrics used in this project

All these metrics measure similarity between texts, but they use different features. In our dataset of 192 pairs of texts, for paired comparisons of each student's / student group's task 1 and task 2 submissions, some measures were overlapping, while others resulted in distinctly different similarity scores (a visual overview is shown in Figure 2).

	jaro_winkler	euclidian	compression	jaccard	2-gram	3-gram	tf_idf	bm25	local-doc2vec	large-doc2vec	gte-large-en-v1.5	mxbai-embed-large-v1	gte-Qwen2-7B-instruct
0	0.608067	0.919131	0.128440	0.097166	0.011278	0.000000	0.137377	0.116308	0.172857	-0.044027	0.249725	0.318392	0.098719
1	0.686340	0.917919	0.244734	0.209205	0.051282	0.007067	0.311979	0.298042	0.431299	0.194896	0.438494	0.521655	0.458529
2	0.793158	0.933650	0.373264	0.350785	0.089362	0.040984	0.379743	0.354800	0.574291	0.328684	0.851557	0.685018	0.908602
3	0.781956	0.920481	0.240404	0.211538	0.037344	0.004049	0.289852	0.279499	0.538125	0.244686	0.649761	0.773335	0.693591
4	0.693090	0.915034	0.213665	0.155172	0.015306	0.000000	0.540820	0.313161	0.527881	0.166789	0.713244	0.845082	0.651382
...	...	...	...	...	...	...	...	...	...	...	...	...	...
187	0.800201	0.945435	0.540732	0.489796	0.356250	0.303030	0.796575	0.632587	0.834689	0.582483	0.704517	0.917733	0.838462
188	0.598217	0.899571	0.159030	0.073171	0.000000	0.000000	0.295039	0.162459	0.374939	0.171111	0.454780	0.776092	0.418852
189	0.550723	0.925926	0.092194	0.047727	0.008791	0.002193	0.100575	0.090311	0.223183	0.081316	0.288520	0.568980	0.281053
190	0.565285	0.917568	0.088283	0.052786	0.002809	0.000000	0.038072	0.036525	-0.066730	-0.055490	0.018647	0.104945	0.041235
191	0.705017	0.916638	0.276786	0.230769	0.067308	0.028037	0.528586	0.438977	0.692706	0.342072	0.708840	0.758882	0.624250

192 rows × 13 columns

Table 2: Snapshot of similarity scores from our student text dataset for paired task 1 – task 2 comparisons

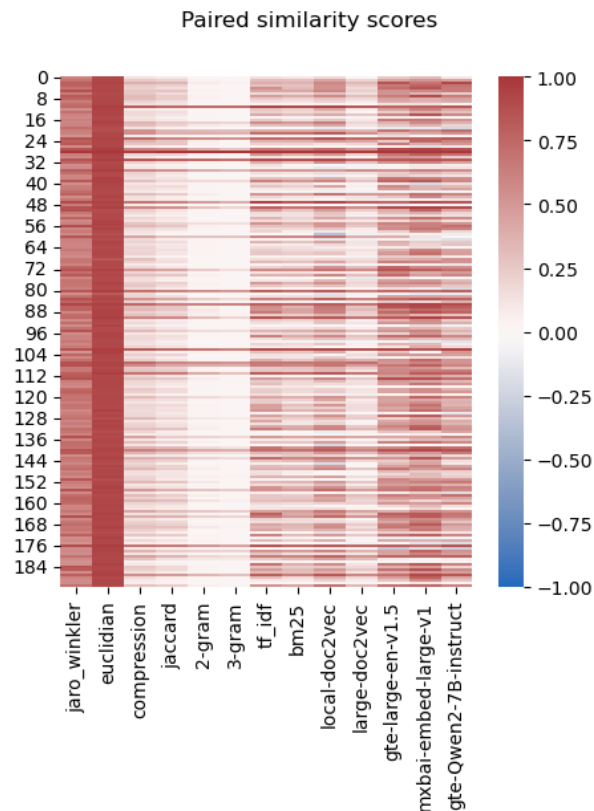


Figure 2: Visual heatmap representation of similarity scores for paired task 1 - task 2 comparisons



I chose these measures for several reasons:

- I wanted to try a range of measures that consider text at different granularity (as explained in the initial proposal) to see how they compare.
- Jaro-Winkler, Euclidian, compression, and Jaccard similarity were conveniently available in public Python libraries: [jellyfish \(jamesturk.github.io\)](https://jamesturk.github.io) and [simphile \(github.com\)](https://github.com).
- Bigram and trigram overlap is a common strategy in Information Retrieval, and it was only a small addition to the Jaccard similarity calculation.
- TF-IDF and BM25 are both classic Information Retrieval approaches for measuring text similarity, although typically they only work well for longer texts. These were both available in the [Gensim: Topic modelling for humans \(radimrehurek.com\)](https://radimrehurek.com) library.
- TF-IDF and Doc2Vec had precedent in previous studies, like Guzman and Li (2023) and Arts, Cassiman, and Hou (2023). An implementation of Doc2Vec was also readily available in the gensim library, but I could not find any up-to-date pretrained Doc2Vec models and therefore trained 2 new models: one trained on our data and another trained on website dataset from Guzman and Li (2023). An interesting consideration here was that many of the business websites in this dataset are protected by copyright. We reused this data under the UK copyright exception for text and data mining for non-commercial research ([Exceptions to copyright - GOV.UK \(www.gov.uk\)](https://www.gov.uk)).
- SBERT models were mentioned as a potential avenue to explore in the future by Guzman and Li (2023). In natural processing research, the Semantic Text Similarity (STS) Benchmark leaderboard is topped by SBERT models. However, the STS benchmark only contains short passages of less than 500 words, so usually such models only work well for shorter texts. Support for SBERT models was available in the [SentenceTransformers Documentation — Sentence Transformers documentation \(sbnet.net\)](https://www.sbert.net) library.
- I picked 3 different open-source SBERT models from the Hugging Face Massive Text Embedding Benchmark (MTEB) Leaderboard ([MTEB Leaderboard - a Hugging Face Space by mteb](https://mteb.com)), which includes the STS Benchmark:
 - At the time of implementing (July 2024), mixedbread-ai/mxbai-embed-large-v1 model was near the top of MTEB leaderboard in embeddings models with less than 1B parameters. I needed to start with smaller models that didn't need much memory and compute to run, so that I could keep testing on Noteable. However, this model had a short maximum sequence length – texts longer than this simply get truncated when computing embeddings.
 - Alibaba-NLP/gte-large-en-v1.5 was another high scoring relatively small model but with a much longer sequence length.
 - Alibaba-NLP/gte-Qwen2-7B-instruct was a much larger model with a sequence length that could accommodate the longest student texts without truncation.

Model	Model size (million parameters)	Memory usage (GB)	Average STS score (Pearson correlation)	Maximum sequence length (tokens)
mixedbread-ai/mxbai-embed-large-v1	335	1.25	85%	512
Alibaba-NLP/gte-large-en-v1.5	434	1.62	81.43%	8192
Alibaba-NLP/gte-Qwen2-7B-instruct	7613	28.36	83.04%	131072

Table 3: SBERT model comparison

Another option could have been to use a closed-source embedding model from OpenAI, but since students used ChatGPT to refine their proposals, I wondered if using a related embedding model might interfere with the results, picking up on features that are specific to OpenAI’s model embeddings, relating to model training data and not the student text themselves. For avoidance of doubt, I chose to use models unrelated to ChatGPT.

1.7. Additional pre-processing and post-processing

Before creating text representations using TF-IDF and BM25 models, I computed a shared vocabulary from all the pre-processed texts and supplemented it with common bigrams and trigrams. Then I filtered out all tokens that occurred in more than 50% of the documents (this eliminated the words “business”, “digital”, “platform”, “transaction”, and “user”). Usually, one would also filter out the least common tokens, but I decided to retain them because unique words add to document originality, which is exactly what we were interested in.

As shown in Table 1, different similarity metrics have different ranges. I considered scaling cosine similarity in 2 ways: either adding 1 and dividing by 2 (which simply compresses the range from [-1,1] to [0,1]) or using the absolute value (negative cosine similarity is meant to indicate opposite meaning, which is still a kind of similarity as similar topics must be involved, unlike values near 0 that indicate no overlap between topics) but decided against either approach in the end as I couldn’t find any clear evidence to support either strategy.

A paragraph from notes about one open-source model (<https://huggingface.co/BAAI/bge-large-en-v1.5>) explains that model similarity scores cannot be taken at face value: “Since we finetune the models by contrastive learning with a temperature of 0.01, the similarity distribution of the current BGE model is about in the interval [0.6, 1]. So, a similarity score greater than 0.5 does not indicate that the two sentences are similar.” While this wasn’t one of the models I used to create embeddings, the same concept still applies to all modern embeddings models, so I decided to use a post-processing technique to rescale the score range. Borrowing the idea from Mu, Bhat, and Viswanath (2017), for embeddings produced by each model, I subtracted the mean embedding for all texts before computing similarity scores. This eliminated the most common vector and extended the range of similarity scores.

I tried including some mock data (a couple of sentences about Teletubbies) to see how they would impact the data and found that they skewed the mean embedding vector. To avoid this, I decided to use a more robust mean average calculation by first trimming the top and bottom 25% values (only leaving the interquartile range) and then calculating the mean.

Further work is still required to investigate whether this is sufficient or if another rescaling approach may be needed to get comparable scores between models.

1.8. Paired similarity scores

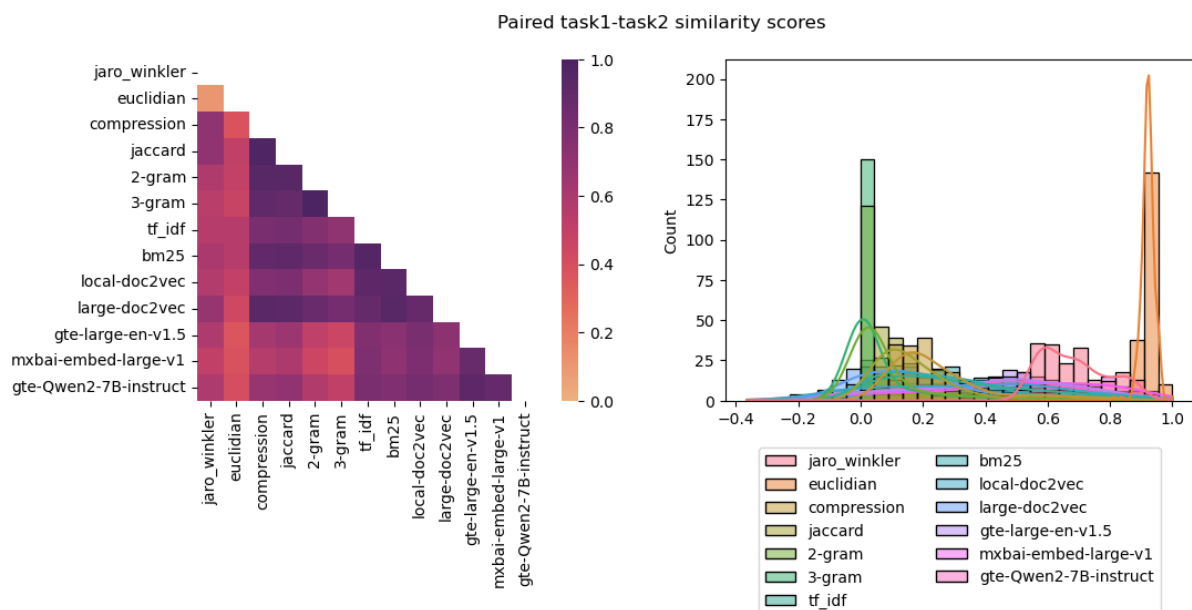


Figure 3: Correlations and distributions of paired task 1 – task 2 similarity scores using different measures

Figure 3 shows a visualisation comparing the correlations between and distributions of the similarity measures. Looking at the correlations, I found 4 distinct groupings of measures:

1. Jaro-Winkler and Euclidian metrics were not very correlated with any of the other measures. Both measures are character-level similarity measures. Since we were not particularly interested in analysing the orthography of students' submissions and these metrics did not seem to relate to any of the other metrics, we decided not to use them.
2. Compression, Jaccard, 2-gram, 3-gram, TF-IDF, BM25, and Doc2Vec similarities were all fairly highly correlated. This suggested that one or two metrics representative of this group could be useful for our final measure.
3. TF-IDF, BM25, Doc2Vec, and contextual embeddings (SBERT) model similarities had medium correlations. This led me to think that if we either choose static or contextual embeddings, we will miss out on information that the other approach captures.
4. SBERT model similarities were highly correlated. Again, this suggested that one or two models representative of this group could be sensible to include.

Another interesting finding is that despite its short (512 token) maximum sequence length, the mx-bai-embed-large-v1 model similarity scores were highly correlated with the other models with much longer maximum sequence length, like gte-Qwen2-7B-instruct. One possible explanation could be that the first 512 tokens of longer texts is enough to form a complete representation of the general meaning of the text. Another explanation could be that the other models, although supporting longer sequences, might not actually have been trained on texts that long, which would lead to them giving little to no attention to later parts of texts during inference (this is supported by the notes on another open-source model ([dunzhang/stella_en_400M_v5 · Hugging Face](#)), which mention “Q: What is the sequence length of models? A: 512 is recommended, in our experiments, almost all models perform poorly on specialized long text retrieval datasets. Besides, the model is trained on datasets of 512 length. This may be an optimization term.”).

If the second explanation is true, a workaround could be to:

1. Split up the texts into chunks smaller than model sequence length
2. Compute similarity between all text chunks
3. Pick max similarity between any chunks in comparison between 2 students

Credit to nreimers on [Sentence transformers for long texts · Issue #1166 · UKPLab/sentence-transformers \(github.com\)](#) for the approach suggestion.

Investigating the possible explanations and implementing the workaround are left up to future work.

1.9. Task similarity scores

Figure 4 shows a visualisation of all the similarity scores. It is organised into 3 columns – the first shows similarities between task 1 texts, the second between task 2 texts, and the third between all texts. Each row of heatmaps is for a different metric: TF-IDF, BM25, small Doc2Vec model, large Doc2Vec model, and the 3 SBERT models.

In the first 2 columns, I could not find any clear patterns, but the third column is interesting. In each triangle heatmap in the third column, the upper part of the triangle shows comparisons between task 1 texts (same as first column), the right part of the triangle shows comparisons between task 2 texts (same as second column), and the lower-left square shows comparisons between task 1 and task 2 texts. With all metrics (and most noticeably with SBERT models), the square is distinct from the rest, with much lower (and often negative) similarity scores, except for a downward diagonal line. The diagonal line represents paired comparisons of task 1 and task 2 texts from the same student / student group. It means that, overall, task 1 texts were dissimilar to task 2 texts, except to their paired ones.

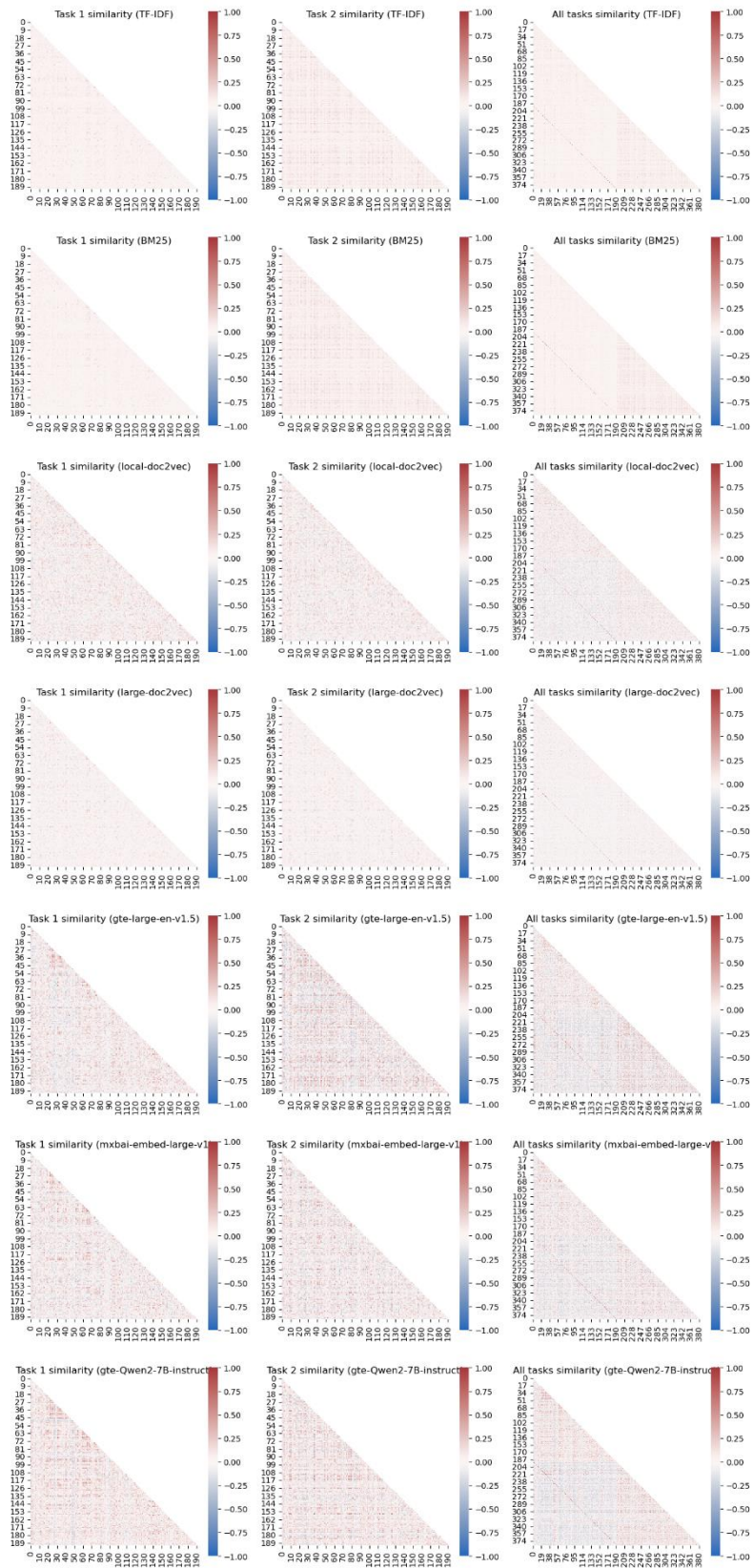


Figure 4: Similarity heatmaps for different similarity metrics

1.10. Originality

Initially, we had decided to use the inverse of mean similarity (distance) from any text to all other texts as our originality metric. However, I realised that this approach would allow a group of closely related texts that are far away from all others to all be “highly original” – but this wouldn’t make sense, since they each would have several close neighbouring texts. Guzman and Li (2023), in their analysis of business website originality, avoided this problem by computing similarity to only the 5 closest texts. This approach would not be suitable for us, as we wanted to incorporate a component of topic uniqueness across all submitted business proposals, but with Jordana, the project host, we decided to build on this by averaging the distance to all texts and the closest 10 texts (the number 10 was chosen arbitrarily).

Considering the pattern of dissimilarity across task 1 and task 2 texts noted in the previous section, I computed originality metrics for task 1 and task 2 texts separately.

Figure 5 shows correlations between average distance computed with different metrics, as well as their distributions, split by task and top 10 / all texts. Notably, there were correlations between TF-IDF and BM25 model scores across all categories, which suggested that only one of them should be incorporated into the final score. In task 2 all text similarity, all scores were highly correlated. In task 1 all text similarity, the Doc2Vec and SBERT models were highly correlated between themselves but not with TF-IDF and BM25. This suggested that it may be a good idea to include one of TF-IDF or BM25 and one of Doc2Vec or SBERT models. For the average distance to closest 10 texts, there was very little correlation between measures (except TF-IDF and BM25).

Combining with the findings from paired similarity scores, I opted to use BM25 and the large Doc2Vec model as the more advanced techniques for static embeddings that were representative of this group and chose to use gte-Qwen2-7B-instruct model scores as this was the most advanced model of the group and representative of the other contextual embeddings models.

The final originality score for a proposal was thus calculated as the average of:

1. Task 1 top 10 texts mean cosine distance
2. Task 1 all texts mean cosine distance
3. Task 2 top 10 texts mean cosine distance
4. Task 2 all texts mean cosine distance

For the models: BM25, large Doc2Vec and gte-Qwen2-7B-instruct.

Being based on cosine distance, this measure has the range [0, 2].

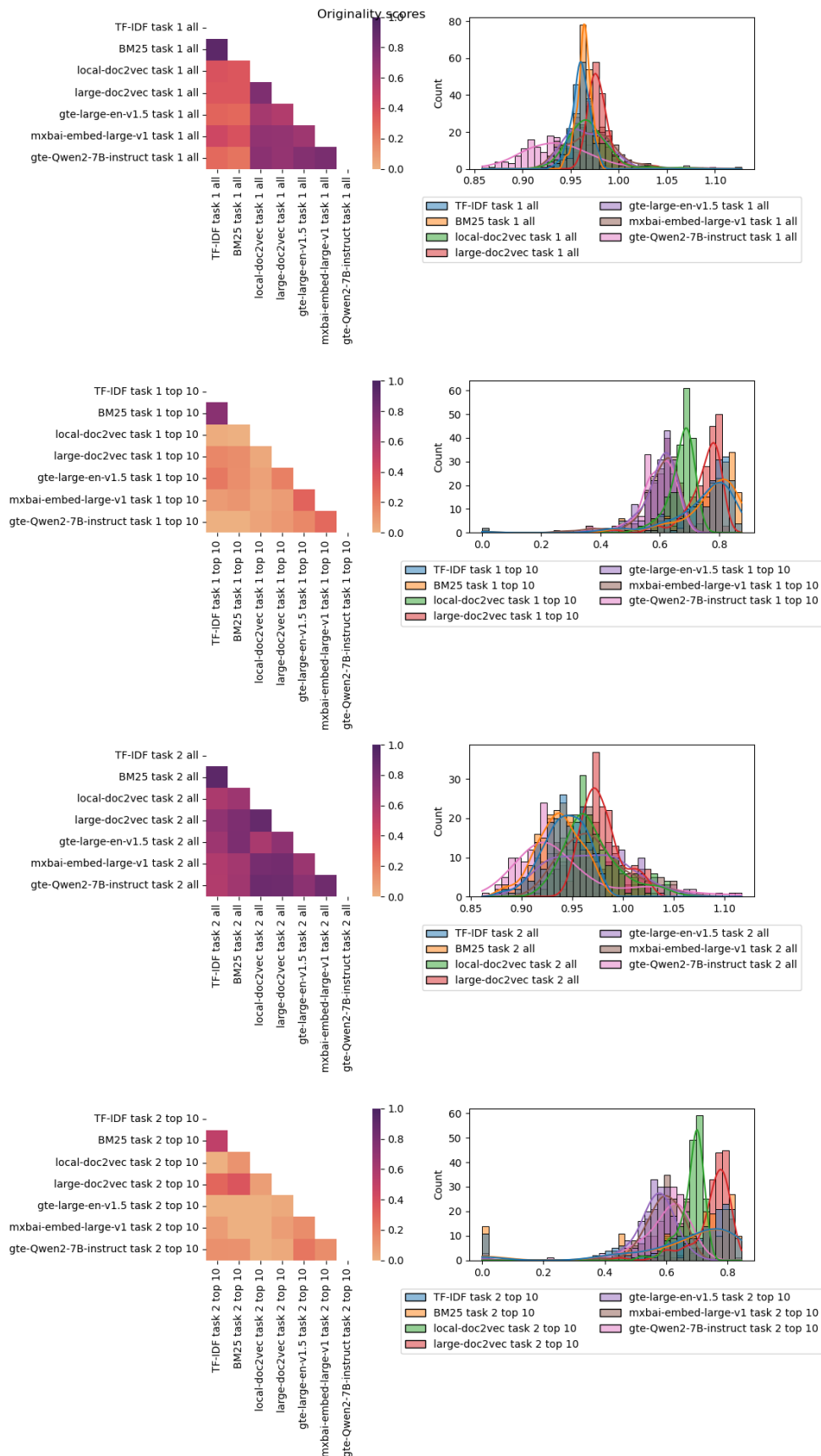


Figure 5: Originality metrics correlations and distributions



2. ACTIVITIES PLAN

Week 1 (29 April – 5 May): First DRS Ambassadors meeting, completing induction checklist, and refreshing compulsory training courses.

Week 2 (6 – 12 May): Initial project meeting, reviewing tasks, estimating potential solutions and how long it might take to implement

Week 3 (13 – 19 May): Finishing background research and writing project proposal document / away on holiday

Week 4 (20 – 26 May): Away on holiday

Week 5 (27 May – 2 June): Implementing similarity measures from project proposal using synthetic data and Noteable

Week 6 (3 – 9 June): Meeting with project host; implementing similarity measures; consulting with colleagues about other potential approaches. DRS mid-programme review.

Week 7 (10 – 16 June): Meeting with Leo (another project intern); correcting some errors in data pre-processing; adding embeddings visualisations

Week 8 (17 – 23 June): Revising which tokens are removed; changing cosine re-scaling; reading about similarity measures and other potential data features to use; project catch-up meeting

Week 9 (24 – 30 June): Investigation of code and dataset from Guzman and Li (2023); Adding new embeddings model to analysis; Minor other improvements; Writing up evaluation strategy

Week 10 (1 – 7 July): Computing originality scores; new visualisation of student text embeddings with UMAP + HDBSCAN + C-TF-IDF; investigation of effects of noisy data on similarity scores

Week 11 (8 – 14 July): Running large embeddings model on ECDF Eddie and Informatics Research Cluster; reading about potential copyright implications of using business website data

Week 12 (15 – 21 July): Project review meeting; training large Doc2Vec model with data from Guzman and Li (2021)

Week 13 (22 – 28 July): Code cleanup; additional documentation; final project report



3. OUTCOMES

3.1. Project Outputs

At the end of the project, I handed over:

- This report which provides an overview and explains my project design decisions
- An Excel spreadsheet with similarity and originality scores for all texts
- Code and instructions to reproduce all the scores
- Accompanying model and embedding files
- Notebook with statistical analysis and interactive visualisations

3.2. Infrastructure for Running Experiments

During this project, 3 university resources were particularly important:

1. Noteable: [Noteable - Computational Notebook Platform for Education \(edina.ac.uk\)](https://edina.ac.uk/noteable/)

I started working using Noteable, as it provided a simple way to quickly write and debug Python code using Jupyter notebooks, while keeping all data on a university service. However, some of the models used for similarity measures required significant compute to train and run, and Noteable only offers limited 4GB RAM and CPU resources.

2. Eddie: [High-performance computing | The University of Edinburgh](https://www.ed.ac.uk/eddie/)

I attempted using Eddie, which all PhD students have access to. It took a little bit of time to learn how to use it but wasn't very complicated and eventually worked well for small tests. However, upon requesting a GPU and 32GB RAM required to compute embeddings using a larger embeddings model, my job got stuck in a waiting queue for 2 days – at this point I cancelled the queued job and gave up on using Eddie.

3. Informatics research cluster: [Research Cluster | Documentation \(ed.ac.uk\)](https://www.ed.ac.uk/informatics/research-cluster/documentation)

Instead, I turned to the informatics research cluster, which, as an informatics PhD student, I also had access to. It took a few days to learn how to use it, as the system is completely different to Eddie, but there was plenty of documentation to help. At the time of this project, the cluster was not very busy, and I could request as many resources as I needed without having to wait long. This was indispensable for training a large Doc2Vec model and computing embeddings using large models.

3.3. Main findings

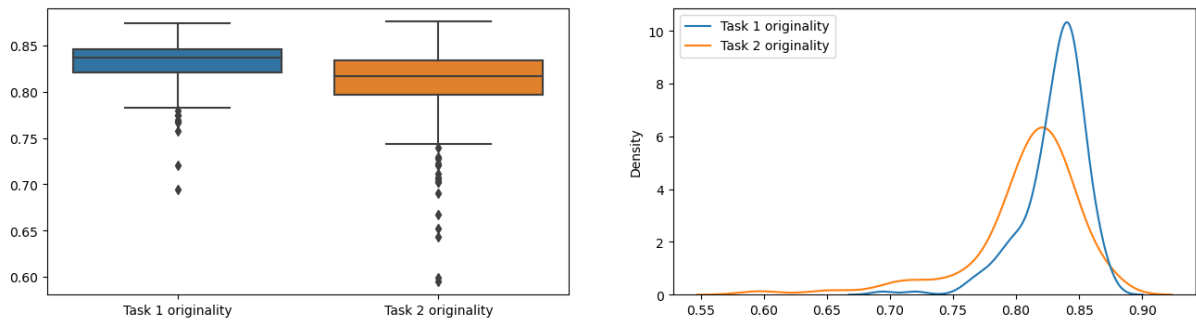


Figure 6: Final originality score distributions by task

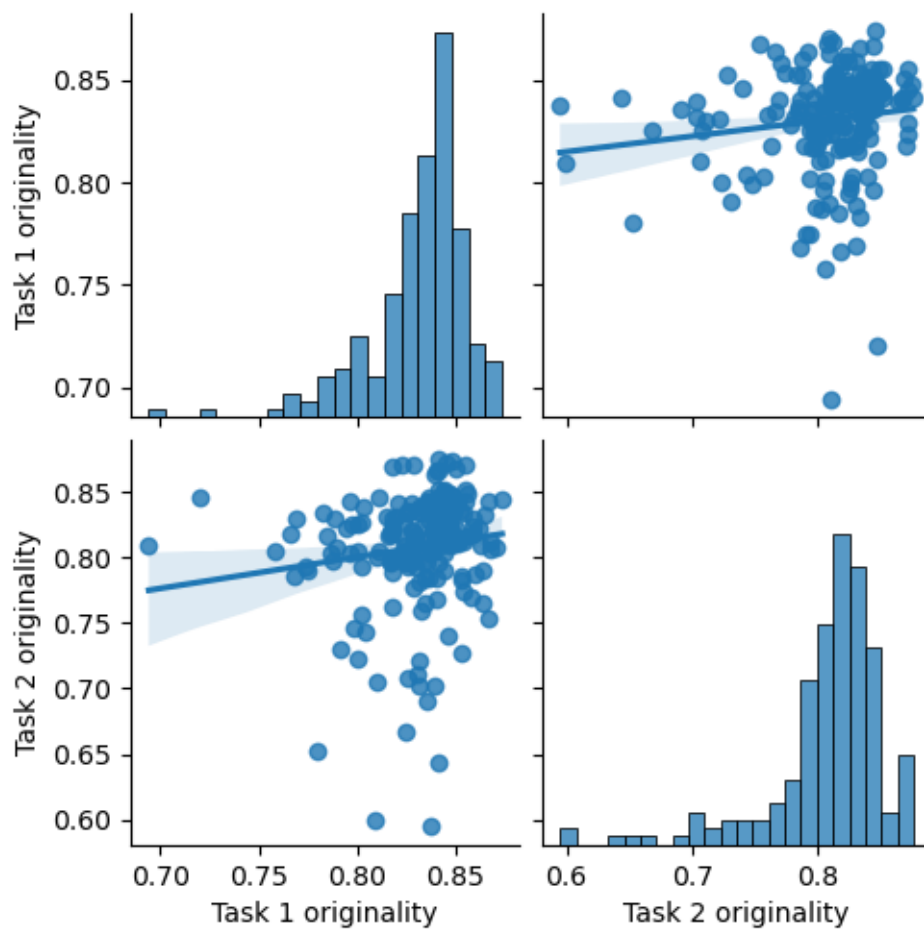


Figure 7: Linear regression of final originality scores by task

I found that there was a statistically significant (according to a Wilcoxon signed-rank test with $p < 1e-9$) but rather small difference between task 1 and task 2 originality, implying that there isn't really any real-world noticeable difference between student-written and ChatGPT-refined business proposals.

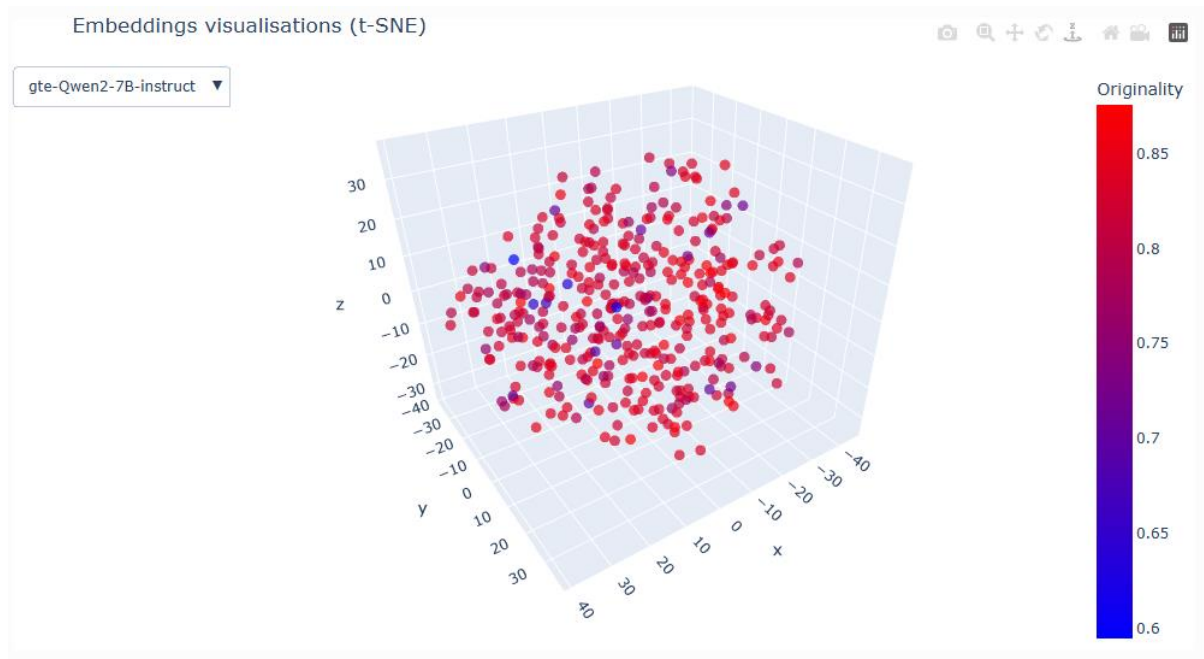


Figure 8: 3D visualisation of gte-Qwen2-7B-instruct embeddings of proposals coloured by originality

With our chosen originality measure, most proposals had moderate originality, and only a few outliers stood out as less original. None of the texts had originality of 1 or higher, which is unsurprising, given that they were all written for the same task.

3.4. Extensions

In an extension to the project, I experimented with several dimensionality reduction and clustering techniques to identify groups of similar texts using all the embedding models. UMAP and HDBSCAN resulted in the most interesting visualisation, which I supplemented with C-TF-IDF to identify topic keywords for each cluster. Examples of this are shown in Figure 9 and Figure 10. The analysis revealed several recurring clusters identified by all models around the themes of carbon emissions and transportation, clothing and fashion, tutoring and learning, and secure payments. Most of the models also separated some unique clusters that were not present in others, like mental health and pets, TikTok influencers, and gaming, although it is not possible to tell if this is due to differences in embeddings, or suitability of the chosen parameters for UMAP and HDBSCAN for the model's embeddings.

I also attempted to use another library, [ddangelov/Top2Vec: Top2Vec learns jointly embedded topic, document and word vectors. \(github.com\)](https://github.com/ddangelov/Top2Vec), to identify document clusters, but found its results less satisfying – it only found 3 generic topics in all the texts (Figure 11).

By introducing some mock data, I investigated the impact of (expected) outliers. With gte-Qwen2-7B-instruct embeddings, some sentences about Teletubbies were clustered together with several other submissions that were not really legitimate business proposals (Figure 12). In future work, distance to mock data could potentially be a way to measure appropriateness

of texts and discard those below a selected threshold. However, this was not the case for embeddings from other models, so the results may be highly dependent on the mock data chosen.

For further project extensions, it would also be interesting to try some hand-engineered features suggested by Guthrie, Guthrie, and Wilks (2008) like text reading difficulty.

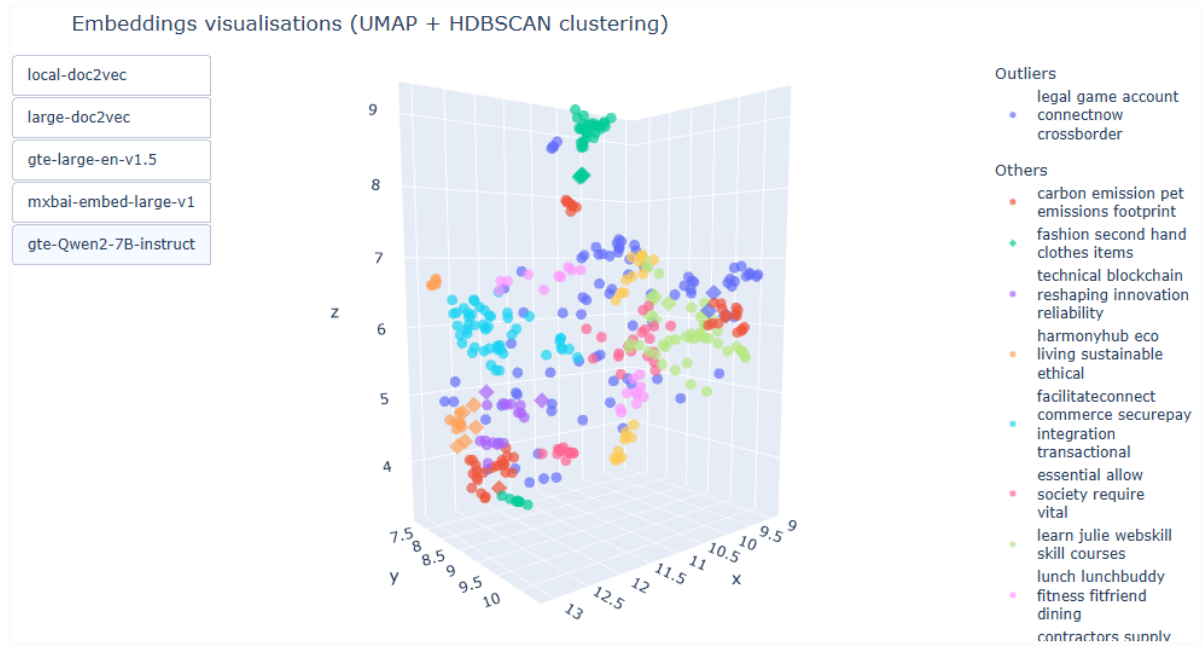


Figure 9: 3D visualisation of gte-Qwen2-7B-instruct embeddings of proposals coloured by topic cluster

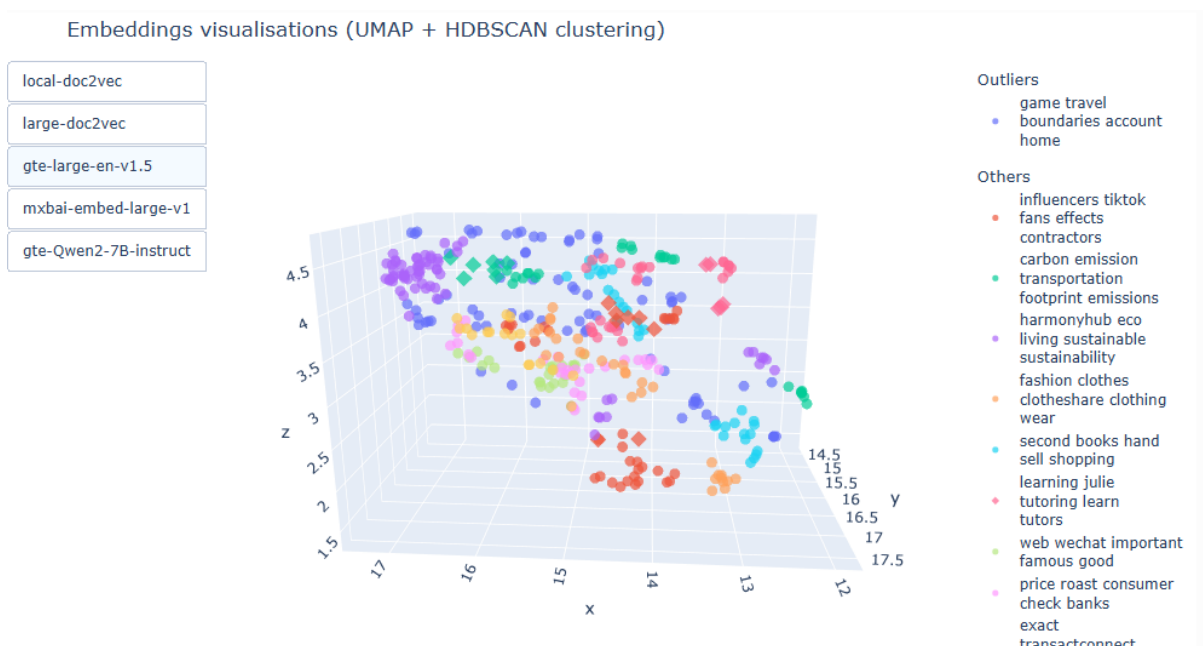


Figure 10: 3D visualisation of gte-large-en-v1.5 embeddings of proposals coloured by topic cluster

Topic 0



Topic 1



Topic 2

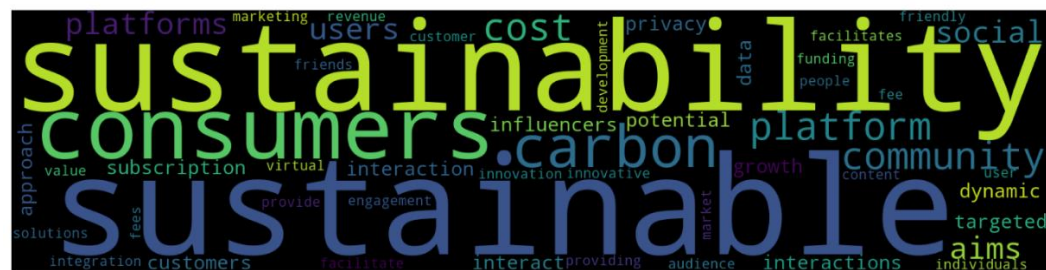


Figure 11: Topic word clouds for topics identified by Top2Vec

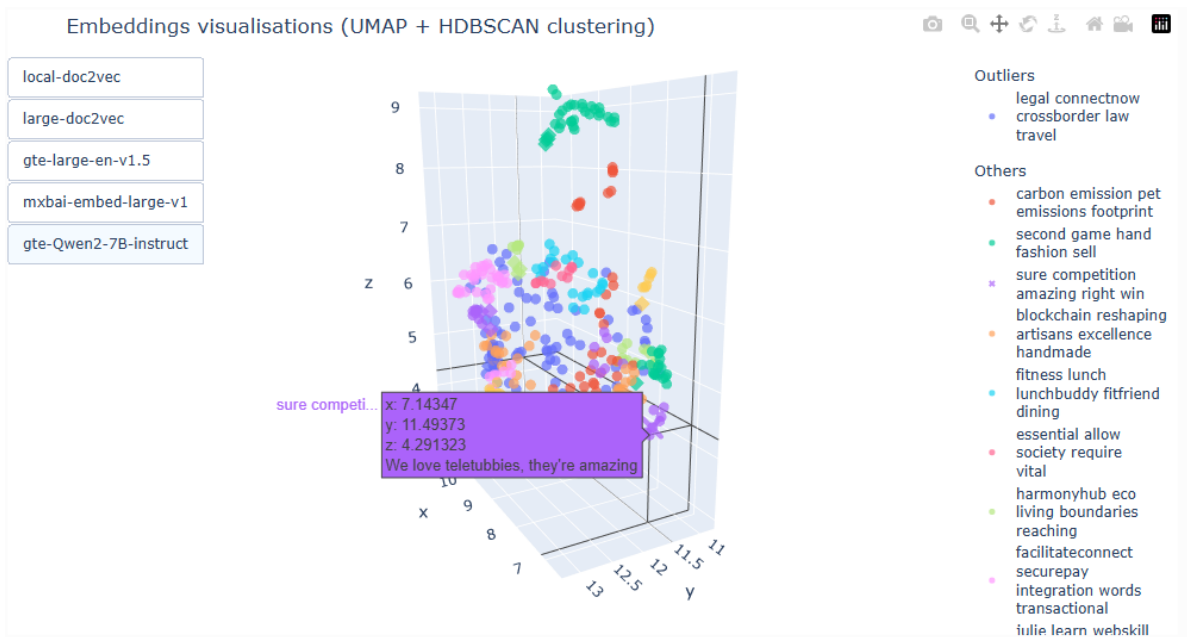


Figure 12: A mock datapoint falls within a cluster of nonsensical proposals



4. CLOSING REMARKS

My analysis and findings will form part of a larger project, involving human marker annotation of originality, appropriateness, and types of interactions with ChatGPT. The human annotators will analyse a larger dataset that includes not only students' submitted proposals, but also full ChatGPT chat histories. As my originality metric choices were mostly based on STS evaluations and correlations between scores but without any source of ground-truth, I am very curious how it will compare to human-labelled originality scores. I am also looking forward to seeing what the remainder of this project will lead to.

Overall, I found this to be a lovely project. I had previously worked with embedding models and information retrieval techniques, but never in this amount of detail, so it was a chance to dive deep and learn a lot more about possible similarity metrics, pros and cons of each, and, more generally, about using open-source models on university compute services.

At each step of this project, I had to think carefully about the design decisions in choosing appropriate processing techniques and metrics. There wasn't much code to write, but lots of conceptual thinking to do. It would have been easy to just pick any distance metric and present it as our originality score, and easy enough to justify any of the choices, but this would have been far less interesting than investigating the whole variety of available options. It was a stimulating intellectual challenge, and I really appreciated the experience. There are still some avenues that I did not have time to fully explore, but I am happy with the results, and feel ready to wrap this up and move on to other projects.

5. FEEDBACK

I had a great time during this year's Digital Research Services Ambassador scheme. I was grateful to be allocated to the project I was most interested in. Besides the technical aspects, I enjoyed the opportunity to network with colleagues from the School of Business. We had related ongoing projects about generative AI in the schools of Business and Informatics, so it was useful to meet, not just for the DRS project, but for these other connections too. The project host I worked with was friendly and helpful, and I would definitely consider working with them again in the future.

My mentor and line manager within Digital Research Services, André, helped connect us for the first meeting and organised weekly ambassador drop-in Q&A meetings. I think it was good to have a point of contact in case there were any issues – but in my case, the only thing I needed to sort out was a period of 1.5 weeks when I was away on holiday. Otherwise, I was happy to take the lead in organising project meetings and monitoring my own progress, so we didn't interact with André much.

As this wasn't my first time working with Digital Research Services, I started out with a good idea of what to expect and what was expected of me. This helped to get started quickly in this



project, taking initiative and not waiting for approval to try things (within our data protection and ethics considerations, of course).

It was good to meet the other Digital Research Services interns. This year we only met a couple of times (admittedly, I didn't put too much effort into organising meetups due to travel and other commitments). Because our projects had little overlap, without these meetings, I would have been completely disconnected from the rest of the work going on in Digital Research Services, so it was nice to meet and catch up.

The timing of the internship was quite good – it had little overlap with the academic year. One of the reasons why I applied for the role is because I work as a teaching assistant throughout most of the year, but usually have less work in May–August, when teaching is on pause.

Overall, I think the DRS Ambassadors scheme was very well organised and I would recommend it for those looking to build up some work experience. It requires applicants to be flexible about what they do, and I would not recommend it for those who have a fixed idea of what they want to work on. It's a good way to meet people from other parts of the university, and therefore great for those in earlier stages of their studies or those intending to stay at the university after studies.

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